Republic of the Philippines

PARTIDO STATE UNIVERSITY

COLLEGE OF ENGINEERING AND COMPUTATIONAL SCIENCES

Department of Computational Sciences | Goa, Camarines Sur

COMPREHENSIVE OPERATIONAL, SYSTEM ADMINISTRATOR, MAINTENANCE & DISASTER RECOVERY MANUAL

"AURORA: Paperless Academic Defense Workflow System for Research, Capstone, Thesis & Dissertation Papers at Partido State University"

Proponents (BSIT 4C)	Gene Vincent D. Mañago • Carol Ann B. Castillo • Rona Mae S. Buenafe • Lara Mae B. Encinas • Leomer Bares
Capstone Adviser & Coord.	Froiland B. Peñalosa
Dept. Chair & Capstone Reader	Kennedy C. Cuya, DIT
Panel of Examiners	Salvador V. Briones II, MAEd (Chair) • Rowell John B. Artiaga, DIT • Michael B. Amata • Froiland B. Peñalosa • Kennedy C. Cuya, DIT
College Dean	Engr. Emelina R. Padayao, SE (College of Engineering and Computational Sciences)

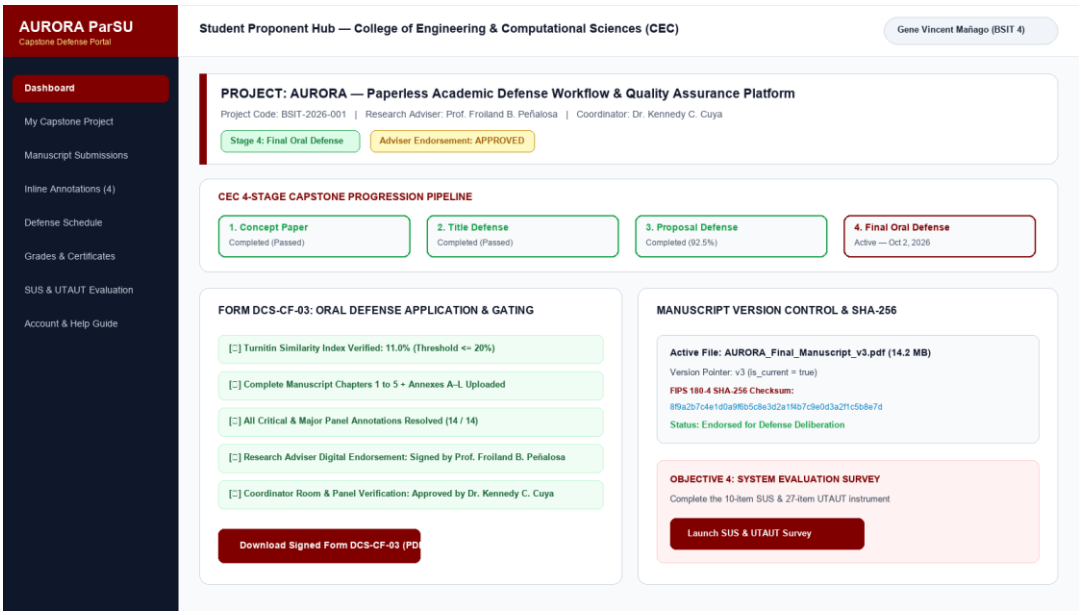


Figure 1. Student Proponent Hub & Multi-Stage Defense Stepper

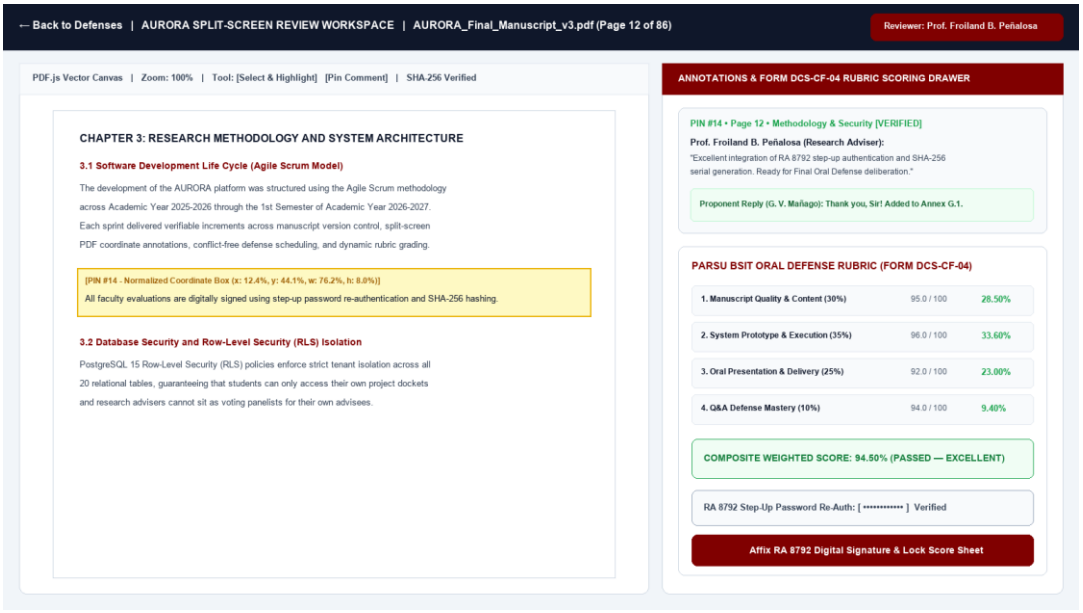


Figure 2. Split-Screen PDF.js Coordinate Annotation & Rubric Workspace

J.1 USER'S GUIDE / ROLE MANUALS

AURORA provides dedicated operational guides for each institutional stakeholder group:

J.1.1 Student Proponent Guide

1. **Accessing the Portal:** Visit `https://aurora-parsu.vercel.app/login` and log in using your institutional email.
2. **Team Formation:** The lead proponent clicks **Create Project**, inputs the approved title, and shares the generated 8-character **Join Code** with teammates.
3. **Manuscript Submission:** Navigate to **Submissions** or **My Project**, drag-and-drop your latest PDF manuscript, and select the target defense stage from your college's customized workflow (4 stages for CEC; 3 customizable stages for other colleges). The system computes its FIPS 180-4 SHA-256 fingerprint and sets status to `under_review` (`pending` adviser endorsement).
4. **Filing Defense Application (DCS-CF-03):** Fill in the requirements checklist (Turnitin score, adviser clearance, preferred defense dates).
5. **Viewing Feedback & Remarks:** Open the **Annotations** view or **Review Workspace** to view adviser and panelist comments. Click on any annotation to jump to the highlighted page segment and reply to notes.
6. **Accessing Defense Outcomes:** Check the **Grades** tab to review composite panel evaluations, download Form DCS-CF-04, and view your verifiable clearance certificate serial number (`AURORA-YYYY-XXXX`).

J.1.2 Research Adviser Guide (e.g., Prof. Froiland B. Peñalosa)

1. **Advisee Directory:** Click **Dashboard** or **Submissions** to view all capstone groups under your supervision.
2. **Dual-Pane Annotation:** Open the project in the **Review Workspace** (`/workspace`). Select text or draw freehand / bounding-box annotations directly on the manuscript PDF. Categorize comments by severity (*Critical*, *Major*, *Minor*) and tag rubric criteria.
3. **Endorsing for Defense:** When satisfied with revisions, click **Approve & Endorse for Defense** (or **Request Revisions**) on the manuscript card. Endorsement automatically unlocks the project for defense scheduling by the Coordinator.

J.1.3 Faculty Panel Evaluator Guide

1. **Defense Calendar:** View your assigned deliberation schedule under **Defenses**.

2. **Live Evaluation Workspace:** During the oral defense, launch the split-screen **Review Workspace** (`/workspace/[projectId]/[stageId]`) to view the PDF manuscript on the left and the interactive rubric scoring drawer on the right.

3. **Scoring Criteria (DCS-CF-04):** Score each dynamic rubric criterion (weights normalized to 100%).

4. **Digital Signature (RA 8792):** Ensure your official signature baseline is registered in **Settings** (`/dashboard/settings`) or draw/upload your signature directly in the signing modal, then click **Finalize & Lock Evaluation**.

J.2 SYSTEM ADMINISTRATOR MANUAL

J.2.1 Defense Coordinator Command Hub

1. **Application & Endorsement Vetting:** Review adviser-endorsed manuscripts and DCS-CF-03 applications.

2. **Defense Scheduling (`/dashboard/defenses/schedule`):**

- **Single Schedule:** Select room, start time, duration, and panel committee. The system automatically blocks room collisions, panelist time clashes, or adviser conflict-of-interest violations.

- **Batch Auto-Scheduler:** Automatically distribute multiple eligible projects across conflict-free time windows and rooms.

3. **Customizable Defense Stages (`/admin/stages`) & Rubrics (`/admin/rubrics`):**

- CEC utilizes the 4-stage progression (*Concept Paper*, *Title Defense*, *Proposal Defense*, *Final Oral Defense*), while other colleges default to the 3 main defense stages (*Title Defense*, *Proposal Defense*, *Final Oral Defense*) with full stage and rubric customization.

J.2.2 College Dean & System Administrator Operations

1. **Executive Analytics (`/dashboard/analytics`):** Monitor college-wide throughput, pass/fail ratios, Objective 2 latency benchmarks, and Objective 4 (SUS & UTAUT) acceptability survey statistics.

2. **User Management & Security (`/admin/users`):** Approve pending registrations, onboard faculty accounts, and manage role assignments (`student`, `adviser`, `panelist`, `coordinator`, `college_dean`, `sys_admin`).

3. **Maintenance, Troubleshooting & Disaster Recovery Console (`/admin/system-health`):** Run live subsystem diagnostics, trigger one-click self-healing maintenance, and export/verify SHA-256 signed institutional database backups.

4. **Audit Ledger (`/admin/audit`):** Inspect the immutable audit trail recording actor email, role, action type, IP address, and state diffs.

J.3 MAINTENANCE & TROUBLESHOOTING GUIDE

J.3.1 Preventive System Maintenance Schedule (SOP)

Frequency	Responsible Role	Target Subsystem	Preventive Maintenance Procedure	Verification Metric

Frequency	Responsible Role	Target Subsystem	Preventive Maintenance Procedure	Verification Metric
Daily	System Admin / Automated Probes	PostgreSQL 15 Pool, RLS & Realtime WSS	Verify database connection pool health, Row-Level Security (RLS) enforcement, automated daily WAL backup completion, and WebSocket latency.	Status = Healthy; Latency < *200 ms*; 0 RLS errors.
Weekly (Fridays)	System Admin / Coordinator	Disaster Recovery Vault & Self-Healing Engine	Click **Export DR Backup Snapshot** in `/admin/system-health`; click **Run Self-Healing Maintenance**; inspect `/admin/audit`.	Signed `.json` backup archived offsite; 100% active pointers.
Monthly	System Administrator	S3 Storage Buckets & SHA-256 Verifier	Audit `manuscripts`, `signatures`, and `certificates` bucket utilization; verify latest backup archive in the **SHA-256 Integrity Verifier**.	Storage < *80%* quota; 100% SHA-256 checksum match.
Semestral Rollover	Coordinator, College Dean & Admin	Workflow Stages, Rubrics & User Directory	Archive completed capstone dockets; review college Defense Stages (`/admin/stages`) and Rubric Templates (`/admin/rubrics`); onboard new faculty.	All college workflows active; rubric weights sum to 100%.

J.3.2 Automated Subsystem Integrity Probes & Self-Healing Maintenance

Accessible via `/admin/system-health` (**Tab 1: Maintenance & Live Diagnostics**), AURORA provides six automated probes and a **Run Self-Healing Maintenance** utility:

1. **PostgreSQL Relational Engine & RLS (`postgres_engine`):** Verifies active connection pooling and Row-Level Security enforcement.
2. **Manuscript Storage & SHA-256 Version Pointers (`manuscript_versions`):** Detects any document where versions exist without an active `is_current = true` flag and automatically promotes the latest version pointer.
3. **RA 8792 Cryptographic Digital Signatures (`digital_signatures`):** Audits all faculty profiles against `user_signatures` and automatically provisions a verifiable SHA-256 baseline signature for any approved faculty account missing one.
4. **College Workflow Templates & Defense Stages (`workflow_stages`):** Verifies ordered defense stages across all university colleges.
5. **Dynamic Rubric Scoring & Consensus Engine (`rubric_engine`):** Confirms active rubric templates have criterion weights summing to 100%.
6. **Security Audit Trail & Non-Repudiation Ledger (`audit_ledger`):** Confirms append-only audit logging across all system modules.

J.3.3 Comprehensive Module Troubleshooting Matrix

Code & Priority	Module	Observed Symptom / Issue	Root Cause	Step-by-Step Resolution
TS-AUTH-01 *(Medium)*	Authentication & RBAC	User logs in but sees *Pending Approval* or restricted navigation.	Account status is `pending` or active role in header Role Switcher is set to another role.	1. Approve the user in `/admin/users`. 2. Use the top header **Role Switcher** pill to switch active role context.
TS-PDF-01 *(High)*	Manuscript PDF & Storage	Uploaded PDF does not appear in Adviser Queue or shows older version.	Upload was interrupted before `is_current` version pointer completed.	1. Open `/admin/system-health` and click **Run Self-Healing Maintenance**. 2. Confirm manuscript status in `/dashboard/submissions` is `under_review`.
TS-WS-01 *(Low)*	Split-Screen Workspace	Realtime annotations do not sync immediately across panelists.	Transient WebSocket (WSS) disconnect or network sleep.	1. Reselect the stage in the workspace header or refresh the page to re-subscribe to the Supabase WAL channel.
TS-SCHED-01 *(Medium)*	Defense Scheduling	*Room Conflict*, *Panelist Clash*, or *Conflict of Interest* error.	Overlapping non-cancelled schedule or adviser selected as voting panelist.	1. Cancel/reschedule postponed sessions in `/dashboard/defenses` first. 2. Select an independent panelist or use **Batch Auto-Schedule**.
TS-RUBRIC-01 *(Medium)*	Rubric Management	Rubric template cannot be saved in `/admin/rubrics`.	Sum of criterion weights does not equal 100%.	1. Adjust criterion weights in the Rubric Builder until total equals 100%, then save.
TS-SIG-01 *(High)*	Digital Signatures & Certificates	Panelist cannot lock evaluation or `/verify/[serial]` fails.	Missing active electronic signature baseline or mistyped certificate serial.	1. Save an Electronic Signature Baseline in `/dashboard/settings` (or run Self-Healing Maintenance). 2. Copy exact serial (`AURORA-YYYY-XXXX`) into `/verify`.

J.4 DISASTER RECOVERY & DATA BACKUP PLAN (DRP / BCP)

To guarantee institutional business continuity and satisfy **ISO/IEC 25010 Recoverability** and **NPC Circular No. 16-01 (Data Privacy Act of 2012)**, AURORA enforces a formal Disaster Recovery and Data Backup Plan accessible at `/admin/system-health` (**Tab 2: Disaster Recovery & Data Backup Plan**).

J.4.1 Service Level Objectives & 3-2-1 Backup Architecture

- **Recovery Time Objective (RTO):** ≤ *4 Hours* for full relational database and object storage restoration (≤ *15 minutes* for cloud application failover).
- **Recovery Point Objective (RPO):** ≤ *24 Hours* via automated daily snapshots and on-demand pre-defense JSON exports (near-zero data loss via continuous Write-Ahead Logging PITR).
- **3-2-1 Backup Rule:**

1. **Tier 1 (Primary Live Cluster):** Managed PostgreSQL 15 cluster with RLS and S3 object storage (`manuscripts`, `signatures`, `certificates`).

2. **Tier 2 (Automated Cloud Daily Snapshots & WAL PITR):** Daily physical snapshots retained for 30 days plus continuous WAL archiving for point-in-time recovery.

3. **Tier 3 (Offsite Cryptographically Signed JSON Cold Archives):** Weekly and pre-defense on-demand snapshots generated via **Export DR Backup Snapshot** (`exportInstitutionalBackupAction`), exporting all 20 relational tables into `AURORA_DR_Backup_YYYY-MM-DD_<sha256>.json` with a verifiable FIPS 180-4 SHA-256 digest.

J.4.2 Catastrophic Disaster Recovery Playbooks

Scenario	Target RTO / RPO	Step-by-Step Recovery & Restoration Procedure
Scenario A: Relational Database Corruption or Accidental Deletion	RTO ≤ *2h* / RPO ≤ *24h*	1. Restrict writes or enable maintenance mode. 2. Trigger Supabase Point-in-Time Recovery (PITR) for cluster rollback, or validate the latest `AURORA_DR_Backup_*.json` in the **SHA-256 Integrity Verifier** (`/admin/system-health`) and restore affected tables. 3. Click **Run Self-Healing Maintenance** to verify foreign-key pointers.
Scenario B: Manuscript PDF Storage Outage or File Tampering	RTO ≤ *3h* / RPO *0h*	1. Compare `document_versions.checksum_sha256` against binary file hashes in the `manuscripts` bucket. 2. Restore missing/altered PDFs from the replicated institutional S3 archive. 3. Re-verify SHA-256 digests before re-opening the workspace.
Scenario C: Compromised Credentials or Auth Lockout	RTO ≤ *30m* / RPO *0h*	1. Set compromised profile status to `suspended` in `/admin/users` (Edge Middleware revokes active sessions immediately). 2. Rotate `SUPABASE_SERVICE_ROLE_KEY` if administrative keys were exposed. 3. Audit `/admin/audit` by `user_email` and `ip_address`.
Scenario D: Cloud Application Host Failover	RTO ≤ *15m* / RPO *0h*	1. Promote the last known-good immutable production deployment in Vercel. 2. Confirm environment variables (`NEXT_PUBLIC_SUPABASE_URL`, `NEXT_PUBLIC_SUPABASE_ANON_KEY`, `SUPABASE_SERVICE_ROLE_KEY`). 3. Click **Re-Run Diagnostics** in `/admin/system-health`.

Compliance Verification: All preventive maintenance procedures, self-healing probes, troubleshooting playbooks, and 3-2-1 disaster recovery workflows are live and operational in `/admin/system-health`.